

New 2023 ERG workflow

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Contents

Overview	2
Requirements.....	2
Getting code from GitHub	3
Selecting the correct folder when using MATLAB	5
STEP1 – Create .JSON from .accdb or .mdb file	6
STEP2 – Convert .JSON file into structure inside a .mat file	9
STEP3 – Exclude repeats from the .mat file.....	10
STEP4 – Exclude flashes and generate average data	11
STEP5 – Export final averaged data to excel.....	14
Scripting reference and other notes.....	15
Data terminology (see formattedMat)	15
formattedMat structure overview.....	16
convertMatToJSON.m.....	18
Opening .fig files programmatically (for figures saved before 1-17-2024).....	18
Note on BASE files.....	18
Looped/Batch variants of scripts	19
Variant scripts (usually _VAR_ in script name)	20

Overview

This workflow provides a new method of exporting ERG data from .mdb or .accdb files into more usable .json, .mat, and .xlsx files. It also creates a standardized data structure within a .mat file that makes any future processing much easier via MATLAB scripting. The following document will step through the process from start to finish.

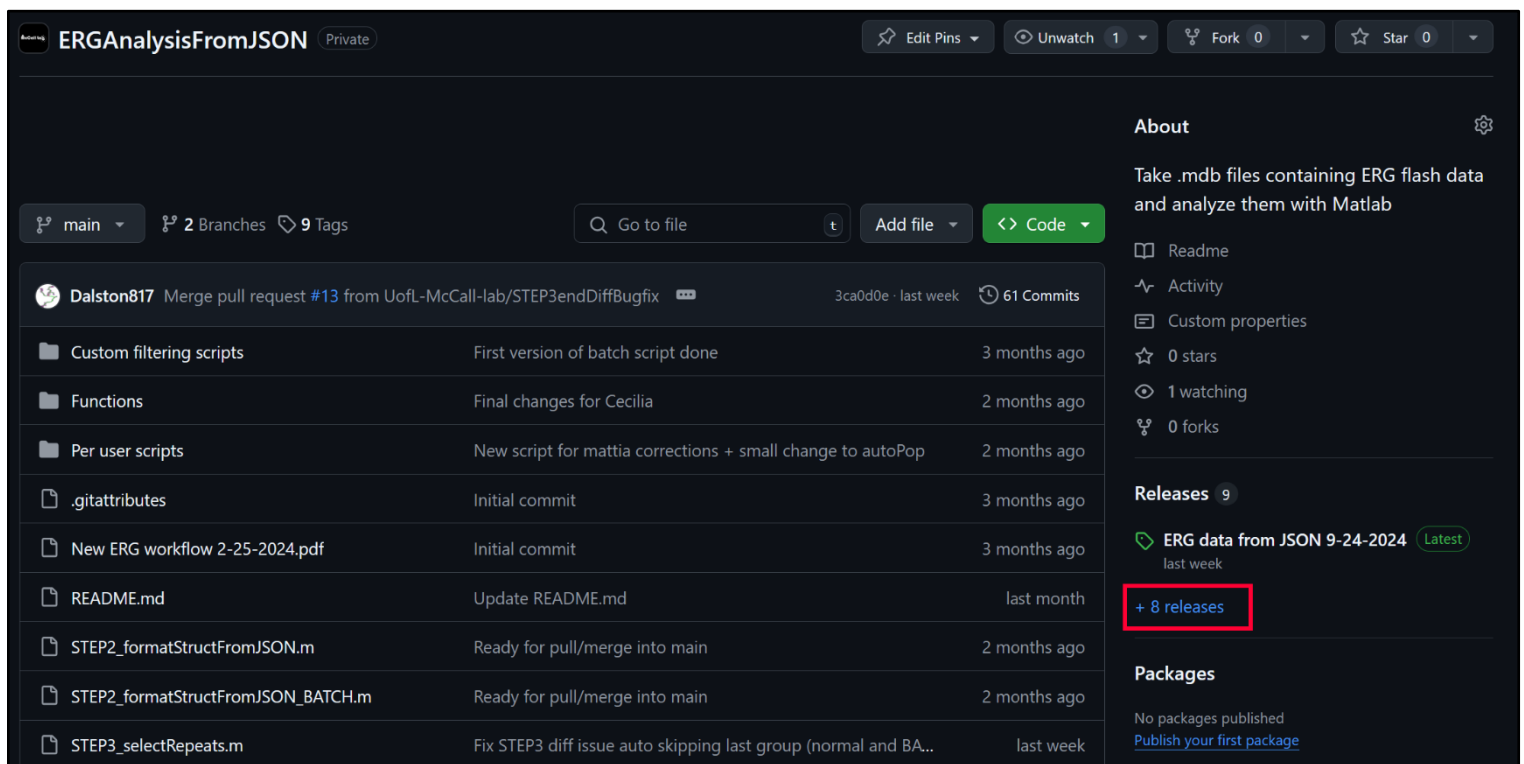
Requirements

- An install of the free community version of dbeaver from this link:
<https://dbeaver.io/download/>
- An install of MATLAB version R2022b or later. No toolboxes should be needed.
- An install of Microsoft Excel.
- Make sure your computer shows the correct date/time (this is saved in STEP4 average editing to keep track of edit history).
- **Optional** – An install of notepad++ (useful to view contents of any JSON files).

Getting code from GitHub

After the initial MATLAB/dbeaver install, you will need to get the most recent release from the McCall lab GitHub:

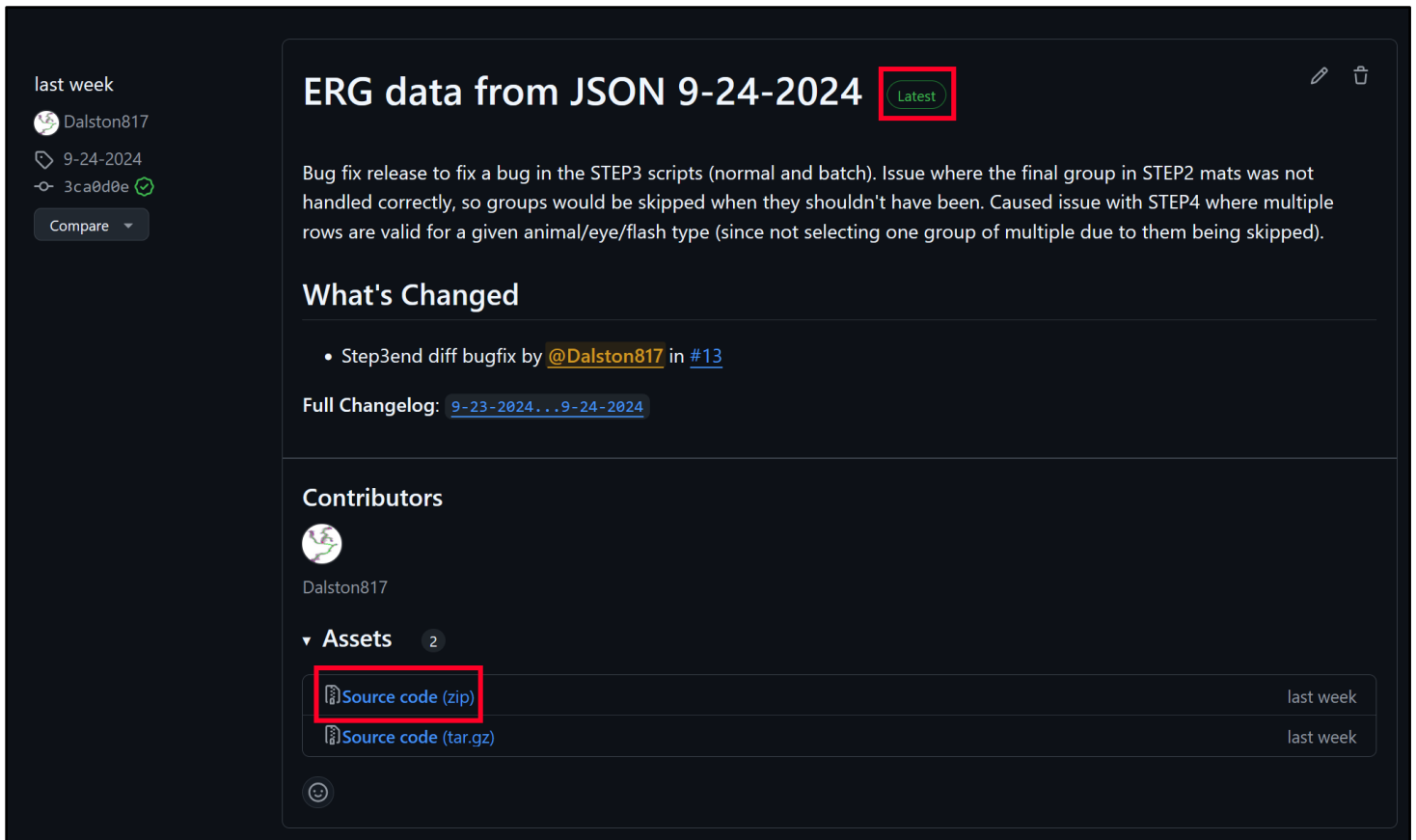
1. Create a GitHub account and set up two factor authentication.
2. Send your username to one of the owners of the McCall lab GitHub. As of 10-1-2024 these are:
 - a. David Alston (david.alston@louisville.edu).
 - b. Archana Jalligampala (archana.jalligampala@louisville.edu).
3. Once you have been added to the McCall lab GitHub, go to the main page of the lab GitHub (<https://github.com/UofL-McCall-lab>).
4. Under the list of repositories, look for “ERGAnalysisFromJSON”. Click this, then click the “Releases” section on the right panel:



The screenshot displays the GitHub repository page for **ERGAnalysisFromJSON**, which is marked as **Private**. The repository is owned by **Dalston817** and has 2 branches and 9 tags. A merge pull request #13 is visible. The file list includes folders for 'Custom filtering scripts', 'Functions', and 'Per user scripts', along with files like '.gitattributes', 'New ERG workflow 2-25-2024.pdf', 'README.md', and several MATLAB scripts. The right-hand sidebar contains an 'About' section, a 'Releases' section with 9 releases, and a 'Packages' section. The 'Releases' section is highlighted with a red box, showing the latest release as 'ERG data from JSON 9-24-2024' with a '+ 8 releases' link.

File/Folder	Description	Time
Custom filtering scripts	First version of batch script done	3 months ago
Functions	Final changes for Cecilia	2 months ago
Per user scripts	New script for mattia corrections + small change to autoPop	2 months ago
.gitattributes	Initial commit	3 months ago
New ERG workflow 2-25-2024.pdf	Initial commit	3 months ago
README.md	Update README.md	last month
STEP2_formatStructFromJSON.m	Ready for pull/merge into main	2 months ago
STEP2_formatStructFromJSON_BATCH.m	Ready for pull/merge into main	2 months ago
STEP3_selectRepeats.m	Fix STEP3 diff issue auto skipping last group (normal and BA...	last week

5. In the releases section you will see the most recent release at the top of the window. Look for the green tag that says “Latest”:

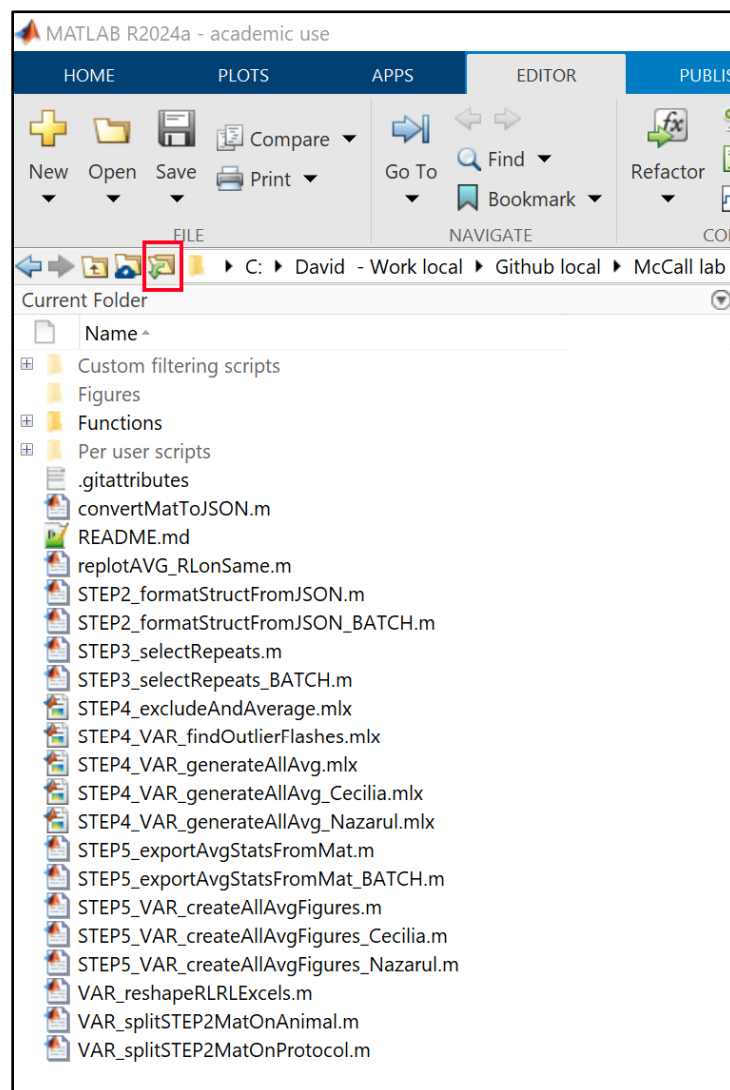


6. Download the zip version of the source code. The tar.gz is also available for Linux/Mac users who prefer a tarball.
7. Move the downloaded file out of your downloads, then unzip it whenever you would like on your computer.
8. You are now ready to use the code. When you start using MATLAB, be sure to switch the current folder in MATLAB to this unzipped folder.

Selecting the correct folder when using MATLAB

Once you have installed MATLAB, you will need to select the correct folder before you can start processing:

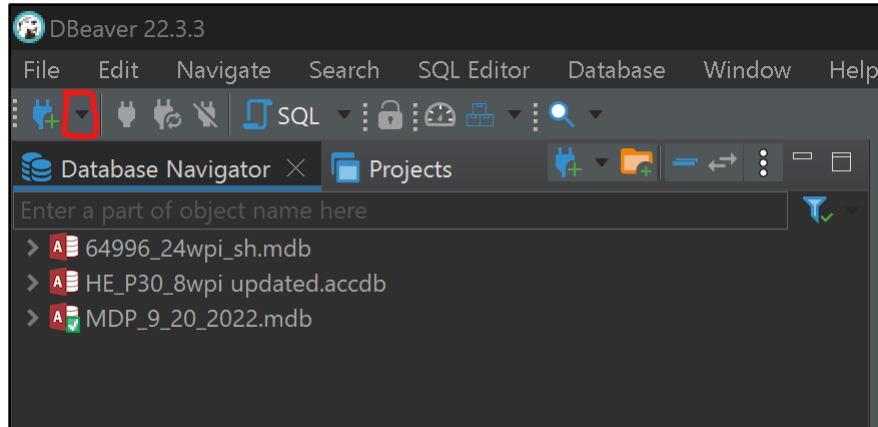
1. Open MATLAB once it is installed.
2. In the top left, look for the icon shown in the image below. Click this button, then select the folder containing the unzipped code (see GitHub section):



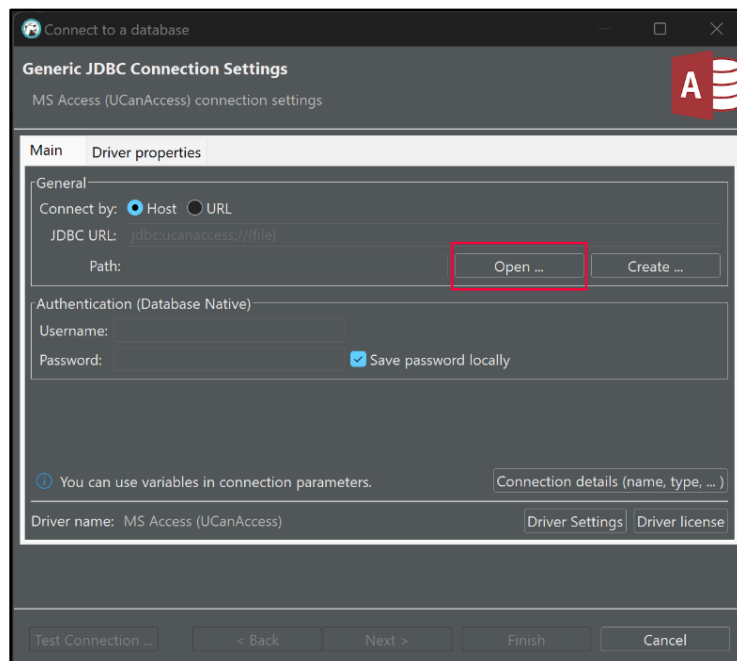
3. You are now ready to start using the scripts to process your ERG data.

STEP1 – Create .JSON from .accdb or .mdb file

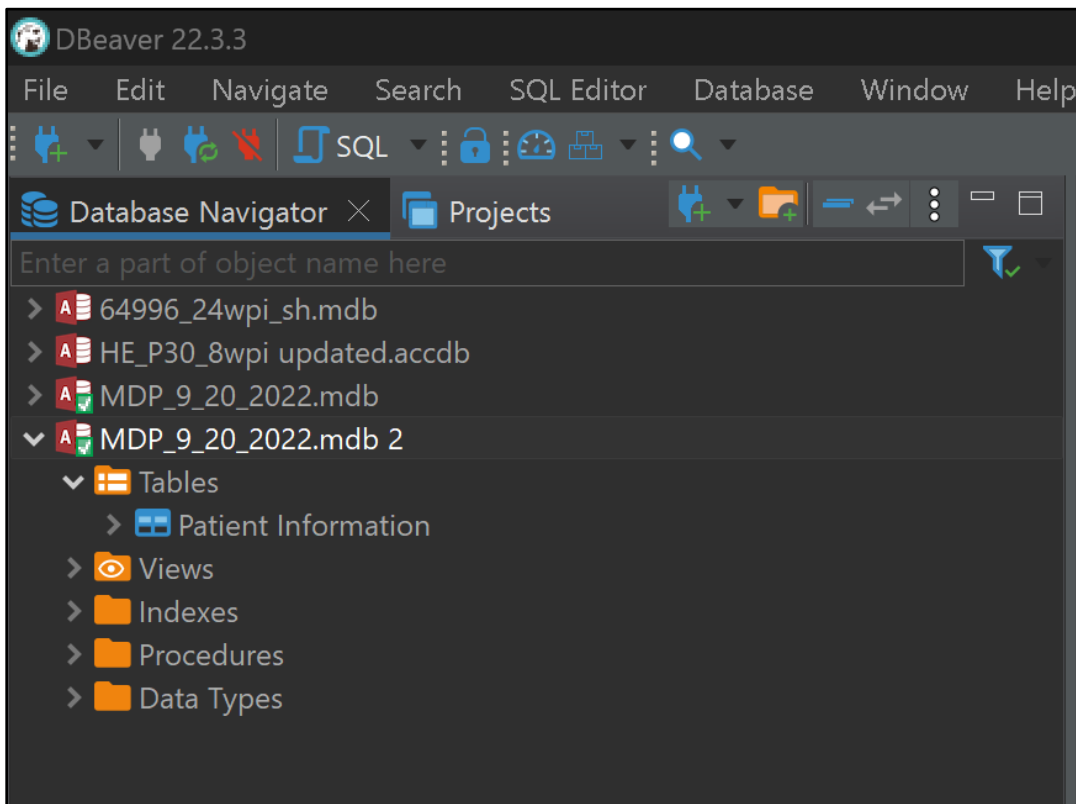
1. Install and then open dbeaver community edition on your computer (from <https://dbeaver.io/download/>).
2. Look for an icon that looks like a power plug under 'File' in the top left corner. Click the dropdown next to that icon:



3. Scroll down to "Other" and find "MS Access (UCanAccess)". Left click this and a new window will open.
4. Left click "Open ..." under the "General" section. When prompted, select the .accdb or .mdb file you want to convert.



5. Click “Finish” and your file should appear in the top left of the main program (look for it by name).
6. In the database navigator, left click the small arrow left of its name to expand it. You should see several new sections: “Tables”, “Views”, etc.
7. Expand the “Tables” section by left clicking the arrow. You should see a new entry called “Patient Information”. This is where all the ERG data is stored:



8. Right click on “Patient Information” then left click “Export Data”. A new data transfer window should open.
9. On the first page of this new window, select JSON as the output format then click the next button near the bottom.
10. Leave “Extraction settings” as their default values and click next.
11. Leave “Format settings” as their default values and click next.

12. In “Output”, look at the “Directory” field. This is showing where the new JSON file will be created. If you want to change this, left click the folder icon and pick a new location.

Otherwise, leave the settings as their defaults and click next.

13. Confirm all the information shown on the final “Confirm” page, then click the “Proceed” button. You should see a new JSON file appear in the chosen directory.

- This JSON file can be examined using a text editor like notepad++.

14. Be sure to rename this file to match the .mdb/.accdb file it came from (since the default file name is “_Patient_Information__XXXX”).

- **This is important to prevent issues with overwriting existing JSON files (accidentally overwriting an existing JSON with new data).**

General tips:

- If you want to export multiple files from dbeaver at once, just shift left click each of their “Patient Information” entries under “Tables” and then right click one of them -> export data.

Then follow the steps above starting at step 9.

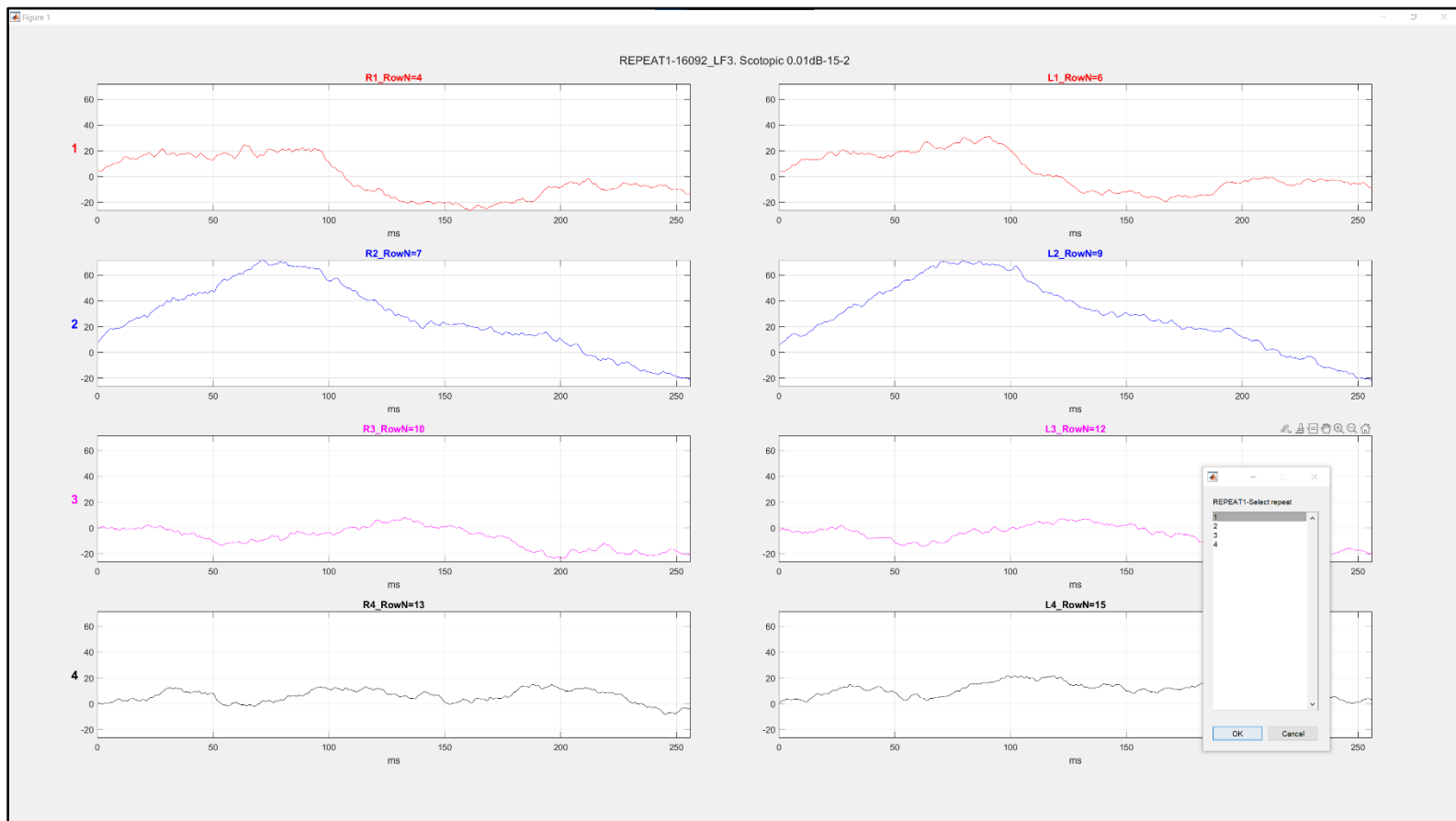
STEP2 – Convert .JSON file into structure inside a .mat file

1. Start MATLAB then open “STEP2_formatStructFromJSON.m” in the editor.
2. Run the script. When prompted, select the .JSON file containing your ERG data.
3. Once finished, the script will prompt you asking where to put the new .mat file and what to call it. By default, it appends “_step2Mat” to the end of the name. Be sure not to change this, as later scripts look for “_step2Mat” to replace that text with something else.

If your data has a non-standard format or structure, you will need to have your own processing scripts after this step. You can use this existing mat, or convert it back into a JSON format if you want to use python etc. Refer to the scripts and feel free to contact David Alston if needed.

STEP3 – Exclude repeats from the .mat file

1. Start MATLAB then open “STEP3_selectRepeats.m” in the editor.
2. Run the code, then in the popup window select a STEP2 mat to process.
3. The program will loop through all the rows in the struct and look for repeats in the waveform data. If repeats are found that need you to select one, a plot of each repeat will appear. A second small window will also popup to select which repeat to use.
4. After selection, it will move to the next repeat (if any exist). You will only see a new window open if a repeat is found AND that repeat requires your selection.
5. Once finished, a prompt will open asking where to save this new mat. It changes “_step2Mat” to “_step3Mat” in the filename to show this step has been done.



STEP4 – Exclude flashes and generate average data

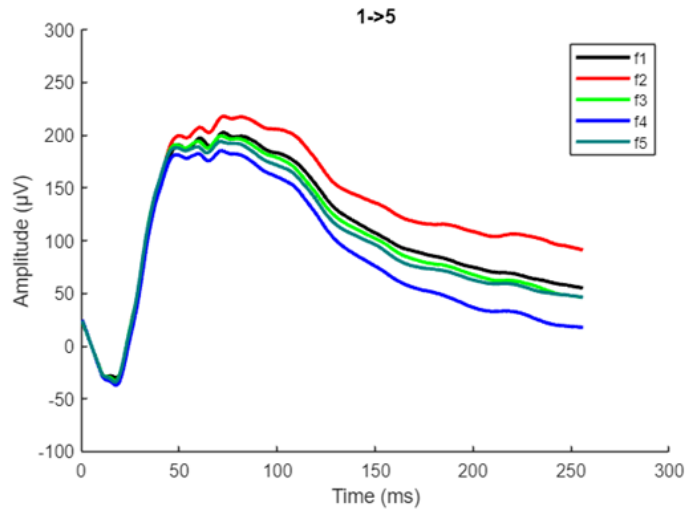
1. Start MATLAB then open “STEP4_excludeAndAverage.mlx” in the editor. Note that this is a live script, so it contains code alongside controls and formatted text.
2. Left click the “Select .mat with ERG data” button.
 - Be sure this has gone through the repeat exclusion script (STEP3). The mat should have “_step3Mat” in the filename.
 - You can also select a “_step4Mat” if you want to continue editing from where you left off (see step 4 in this list).
3. Use the controls to select, view, end edit exclusion/average data for each row in the struct:
 - When you press “Show indiv/avg flashes and exclude”, all the flashes are shown plus the average flash with exclusion applied for the chosen row in formattedMat.
 - This also creates/overwrites .avgStats for this row in formattedMat. See the MATLAB workspace for this variable.
 - **Note that having .formattedMat open for viewing during this processing can slow down MATLAB and make opening new figures slow.**
 - Both plots are automatically saved to the “Figures” folder. If the file for a figure already exists, it will be overwritten.
 - Be sure MATLAB is using the main code folder as its current folder when processing. Otherwise, the code won’t know where to put these figures.

- If you want to change the flash exclusion, just change the entries in “flashesToExclude” then click the “Show indiv/avg flashes and exclude” button again. This will overwrite any existing average data/exclusion information for this row in formattedMat. It will also overwrite the .fig files (if they exist).
 - Original/pre-exclusion flash data is never lost and is reloaded every time you click this button.
 - Leave one space between each flash number. So, to exclude flashes 1 and 10 put 1 10 in the flashesToExclude control.
 - The a-Wave/b-Wave text overlap shown in the image below is a viewing artifact. If you pop out and maximize the figure, the text will be aligned correctly.
 - If you want to remove oscillation in your waveform, try using a larger smoothing window between 50 and 100.
4. When you are finished or want to continue editing later, press the “Save formatted mat to STEP4 mat” button at the very bottom of the script.
- You do not have to process all the rows before moving to the STEP5 script. Any formattedMat rows that are not edited to have .avgStats will be skipped by the STEP5 script.
 - If you just want to view the existing data in the mat (rather than editing it), do not use this script since it edits the formattedMat structure in the workspace every time the “Show indiv/avg flashes and exclude” button is pressed. Instead, drag the .mat file into the workspace and double left click “formattedMat”. Then you can scroll to the right to view average/exclusion data.

Show and exclude individual flashes, then show average

Select animal: 6841-Tg-24wpi
 Select flash type: Scotopic 10dB
☐ If checked, use L eye. Otherwise right
 smoothMethod: sgolay
 smoothWindow/dx: 40
 sgolayDegree: 1
 flashesToExclude: 1
 Show indiv/avg flashes and exclude

6841-Tg-24wpi-R-Scotopic 10dB (5 flashes)-1

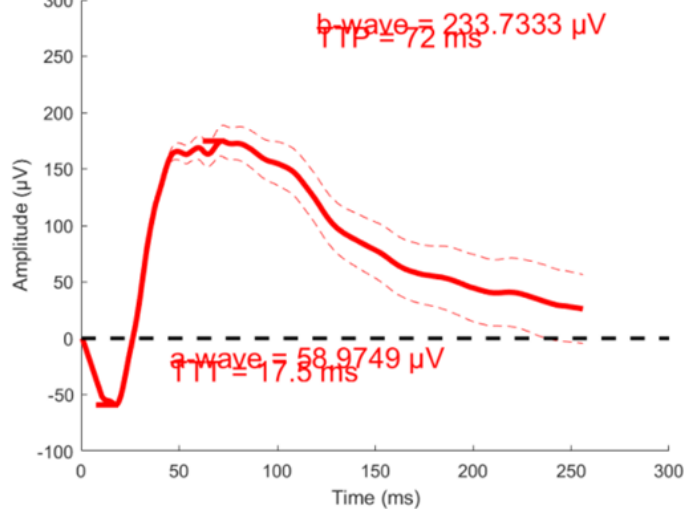


^^^^Saved individual waveforms figure to file = Figures\6841-Tg-24wpi-R-Scotopic 10dB (5 flashes).fig

excluded flash 1

^^^^Saved average waveform figure to file = Figures\AVG-6841-Tg-24wpi-R-Scotopic 10dB (5 flashes).fig

AVG-6841-Tg-24wpi-R-Scotopic 10dB (5 flashes)



formattedMat row 76 updated in workspace with a/b wave, exclusion info, averaging info, etc

STEP5 – Export final averaged data to excel

1. Start MATLAB then open “STEP5_exportAvgStatsFromMat.m” in the editor.
2. Run the script, then select a .mat that has at least one row with .avgStats populated (from STEP4 script).
 - a. Pressing “Show indiv/avg flashes and exclude” in “STEP4_excludeAndAverage.mlx” populates this field with a/b-Wave information (TTP etc.).
 - i. If no average data exists in anywhere in the chosen mat, an error will occur:
Unrecognized field name “avgStats”.
 - ii. Only the rows that have .avgStats populated will be exported to excel.
3. When finished, it will pop up a prompt asking where to put the new excel file. Once you confirm this dialog the new excel file will be created.

Scripting reference and other notes

Data terminology (see formattedMat)

- Group = A group of three rows in the formatted .mat structure. For our data, one group consists of “R”, “[]”, and “L” data.
- Set = Several groups from the same data. Used to indicate repeats (two or more groups from the same data. For example:
 - Set 1 (no repeats)
 - Group 1
 - Row 1 (“R”). See .Wavelabel in the .mat
 - Row 2 (“[]”)
 - Row 3 (“L”)
 - Set 2 (has one repeat)
 - Group 1
 - Row 1
 - Row 2
 - Row 3
 - Group 2
 - Row 4
 - Row 5
 - Row 6

formattedMat structure overview

The initial struct is created during the STEP2 script. All the initial struct variable names match the names that are in the original file, plus new entries added for ease of scripting:

- .groupStartSearch, .toUse, .remReason, .indivFlash, and .avgFlash = Added by formatStructFromJSON.m. Used to make later scripting easier.
 - groupStartSearch = True or false. True indicates the start of a group.
 - toUse = True or false. Allows later scripts to skip over ones that have toUse = false.
 - remReason = If .toUse is set to false, this will contain the reason:
 - AUTOCLEARMID = formatStructFromJSON.m will set this for any rows that have '[]' as their Wavelabel.
 - IDENTICALGROUPS = Set by selectRepeats.mlx automatically if the script detects at least one repeat AND all the repeats in a set have identical waveform data. In this case, the first repeat is automatically selected, and the later ones are edited to set their .toUse = false and .remReason = IDENTICALGROUPS.
 - USERREMOVED = Set by selectRepeats.mlx after the user selects a repeat to use. All other repeats have toUse set to false and USERREMOVED set as their .remReason.
 - .indivFlash has the raw .Data split into individual flashes (one per column).
 - .avgFlash has the mean flash across all flashes for this row (no other processing).

- .smoothBaseCorrFlashes, .baseline, .excludedFlashes, .lastAvgEdit_PC, .lastAvgEdit_Time, and .avgStats = Added and set by STEP4_excludeAndAverage.mlx (STEP4 script).
 - .smoothBaseCorrFlashes = As .indivFlash, but with smoothing and baseline correction applied.
 - .baseline = The baseline for this row. The width of the baseline is defined per type inside the selYLimits_F.m file.
 - .excludedFlashes = Which flashes were excluded (if any). As integers.
 - .lastAvgEdit_PC = Name of PC that did the last average edit. As string.
 - As returned by [~, pcName] = system('hostname');
 - .lastAvgEdit_Time = datetime when last average edit was made on the PC above.
 - .avgStats = All the stats for the final average flash. Calculated by analyzeAverage_F.

Contains:

- finalAverage = Final average flash with baseline correction, smoothing, and re-centering at zero applied.
- minVal and maxVal as [minX, minY] and [maxX, maxY].
- aWave and bWave amplitudes in microvolts.
- Time to peak and time to trough (TTP and TTT) in milliseconds.
- smoothMethod and smoothWindow = Smoothing used and the window size in indices.
- searchWindows_ms and searchWindows_rowTypes = Info about the search windows (per flash type) used when finding the min and max values for aWave and bWave.
- baselineW = The width of the baseline in indices.

convertMatToJSON.m

This script demos how to export a mat to a json file. Use this if you want to use something other than MATLAB to do your scripting/analysis (like python).

- This script expects a .mat containing a single struct (struct called formattedMat for our data).

However, it can be adapted to almost any MATLAB data with minor changes.

Opening .fig files programmatically (for figures saved before 1-17-2024)

To open figure from .fig files:

1. Change the current directory to the one containing the figures.
2. In the command window, enter `openfig('myFigureName.fig', 'visible')` then hit enter.
3. Your figure should pop up in a new window.

This is required for older figures due to an issue where .fig files would be saved from .mlx scripts with the 'visible' property set to 'off'. All the figure data is there, but when opened it is invisible.

Note on BASE files

Certain older files will not plot their flash waveforms with the correct Y limits when run through this system (BASE files for example). See the `setYLimits_F.m` function to see where the Y limits are set. If desired, you can manually change the y limits of a figure using the command window. Open the figure and make sure it is selected. Then go to the command window and type `ylim([yMin yMax])`, where yMin and yMax are your new y limits.

Looped/Batch variants of scripts

Batch versions of several scripts can be found in the main code folder. These modified versions process all the files in a folder into their next state (STEP2->STEP3 for example). They all place their output data into the same folder as the input data. For example, the STEP2 batch script puts all the STEP3 mats it creates into the initial folder of STEP2 mats you selected.

- The batch version of the STEP2 script should be fully automatic, no user input is necessary.
 - Be sure to check for any error messages printed during processing.
- The batch version of the STEP3 script is merely a way to automate stepping through multiple STEP2 mats; it still requires user input for each .mat.
- A looped version of the normal STEP4 script was not created since it involves a lot of user input (selecting flashes).

Variant scripts (usually _VAR_ in script name)

Several variant scripts have been written for specific users' data or to perform specific tasks. A few will be briefly described here:

- STEP4_VAR_generateAllAvg.mlx
 - Used to auto generate all average stats without any single flash exclusion.
Useful if your data is only average data, in which case going one set of flashes at a time is not needed.
- STEP5_VAR_createAllAvgFigures.m
 - Used to auto create all the raw/average flash figures. Exports all data that has .avgData in the struct. If you used STEP4_VAR_generateAllAvg, all valid rows will have .avgData. If you used STEP4_excludeAndAvg, only the rows you edited will have .avgData.
- VAR_splitSTEP2MatOnAnimal and VAR_splitSTEP2MatOnProtocol.m
 - Allows you to split a STEP2 mat into individual STEP2 mats per animal or per protocol. Can be useful to help organize the data and keeps individual files smaller.
- replotAVG_RLonSame.m
 - Takes a folder containing paired R and L figures and combines each pair into a single figure with R and L in the same plot. Be sure the folder you select has an even number of .fig files since this script expects paired R and L figures. Also be sure you have not renamed the figures since this script uses the standardized naming from my code to figure out which figures are paired.